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1. Broad Area of Work

Artificial Intelligence is transforming aviation maintenance and operational reliability through intelligent automation, predictive analytics, and decision-support systems. Aircraft Technical Logbooks (Techlogs) contain large volumes of unstructured defect narratives, maintenance actions, and operational observations which are critical for troubleshooting and maintenance decision-making. However, extracting actionable intelligence from these heterogeneous data sources remains a significant challenge.

The proposed dissertation focuses on developing an Agentic Multimodal AI Framework for aircraft techlog intelligence by integrating:

- Retrieval-Augmented Generation (RAG)
- Large Language Models (LLMs)
- Predictive Analytics
- Multimodal Learning
- Conversational AI
- Safety-Constrained Decision Support
- Explainable AI and Auditability

The framework aims to support aircraft maintenance engineers and operational teams by providing evidence-grounded recommendations, intelligent defect triage, recurring defect prediction, and explainable maintenance insights.

The research project covers the following academic and technical domains:

- Natural Language Processing (NLP)
- Conversational AI
- Deep Learning
- Multimodal AI Systems
- Retrieval-Augmented Generation (RAG)
- Predictive Modeling
- Reinforcement Learning Concepts
- Graph-based Intelligence
- AI Safety and Explainability
- ML System Optimization

2. Background

Aircraft maintenance operations generate substantial volumes of operational and maintenance-related information through electronic Technical Logbooks (eTechlogs), maintenance records, engineering orders, and operational advisories. These records are typically distributed across multiple systems and consist largely of unstructured or semi-structured textual information.

In airline maintenance operations, engineers frequently analyze recurring defects, maintenance history, Minimum Equipment List (MEL) references, troubleshooting procedures, and operational advisories under strict operational timelines. Existing workflows are heavily dependent on manual interpretation, prior experience, and fragmented information retrieval mechanisms, which can increase cognitive workload and response times.

The current work environment within the airline industry increasingly emphasizes:

- Predictive maintenance
- Operational reliability
- AI-assisted troubleshooting
- Safety compliance
- Intelligent decision-support systems

The proposed dissertation is highly relevant to the current organizational environment because it aims to design an intelligent AI-based maintenance copilot capable of:

Retrieving evidence-grounded maintenance information

- Assisting in defect triage
- Identifying recurring defect patterns
- Providing explainable recommendations
- Supporting maintenance engineers with contextual operational intelligence

The project aligns with ongoing industry trends involving digital transformation, AI adoption in aviation maintenance, and intelligent operational analytics.

3. Objectives

The primary objectives of the proposed dissertation are as follows:

1. To study existing AI-based approaches for aircraft maintenance intelligence, conversational systems, predictive maintenance, and Retrieval-Augmented Generation (RAG).
2. To design and develop an Agentic Multimodal AI Framework capable of analyzing aircraft techlog data and maintenance-related information through multiple specialized AI agents.
3. To implement a Retrieval-Augmented Generation (RAG) based conversational assistant (Techlog Copilot) capable of retrieving and citing evidence from maintenance knowledge sources including MEL/CDL extracts, manuals, engineering orders, and prior techlog cases.
4. To develop predictive models capable of identifying recurring defect patterns and generating early-warning indicators using fused textual defect narratives and simulated operational health signals.
5. To design safety-constrained decision-support mechanisms incorporating confidence scoring, abstention logic, escalation handling, and audit logging aligned to aviation AI governance standards.
6. To evaluate the effectiveness of multimodal AI techniques combining textual defect narratives, structured maintenance metadata, and simulated operational signals against single-modality baselines.
7. To study the feasibility of integrating explainable and evidence-grounded AI systems within aviation maintenance workflows in compliance with EASA Level 1/Level 2 AI guidance.
8. To analyze system performance using metrics such as retrieval accuracy , hallucination rate reduction, defect prediction performance, response latency and audit completeness.

4. Scope of Work

The scope of the proposed dissertation includes the independent design and development of an AI-driven prototype system for aircraft techlog intelligence and maintenance decision support. This dissertation is intended for academic and research purposes and does not constitute a production deployment.

The scope includes:

Literature Review on: - Conversational AI and Retrieval-Augmented Generation - Predictive Maintenance and Aircraft Health Monitoring - Multimodal Learning and Foundation Models - Agentic AI Systems and Multi-Agent Orchestration - Explainable AI and Aviation AI Safety Governance

Dataset Creation and Preprocessing using: - Public industrial defect datasets - Synthetic techlog datasets - Publicly available aviation advisories and maintenance-related textual resources

Design and Implementation of: - RAG pipeline - Vector database integration - Embedding generation - Conversational AI interfaces - Predictive modeling components - Evidence-grounded recommendation mechanisms

Development of Multimodal Processing Pipelines combining: - Defect narratives and techlog text - Structured maintenance metadata - Simulated operational/health event streams

Implementation of Safety and Governance Mechanisms including: - Confidence thresholds - Abstention handling - Evidence citation - Audit logging

Experimental Evaluation and Benchmarking of: - Retrieval performance (Recall@k, nDCG) - Defect prediction accuracy (AUROC, AUPRC, F1) - Hallucination rate reduction vs parametric-only LLM baselines - Recommendation quality (SME-graded task success rate) - System latency (p95) and audit completeness

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	05 May 2026 – 10 May 2026	Literature Review and prepare Dissertation Outline
Design & Development	17 May 2026 – 10 July 2026	Design & Development Activity
Testing	11 July 2026 – 06 Aug 2026	Software Testing, User Evaluation & Conclusion
Dissertation Review	07 Aug 2026 - 13 Aug 2026	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	14 Aug 2026 - 19 Aug 2026	Final Review and submission of Dissertation

6. Literature References

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